AI Research Landscape

Structured Taxonomy of Capability, Alignment, and Mechanistic Interpretability Research

Research Outline

I. Artificial Intelligence Research

Capability Research

• Foundation Models: Architectures, Scaling Laws, Training Methods, Data Curation, Multimodal Systems

• Reasoning Systems: Chain-of-Thought, Search, Tool Use, Planning, Agents

• Reinforcement Learning: RLHF, RLAIF, Online RL, Self-Play

• Efficiency Research: Quantization, Distillation, Inference Optimization, Hardware–Software Co-Design

Evaluation and Measurement

• Benchmarking

• Robustness Testing

• Capability Forecasting

Alignment and Safety

• Preference Learning

• Control Methods

• Governance

• Transparency and Interpretability

AI Science

• Learning Theory

• Optimization Theory

• Scaling Laws

• Representation Theory

• Mechanistic Interpretability

II. Interpretability Research

Behavioral Interpretability

• Black-Box Analysis

• Behavioral Probing

Representation Analysis

• Information Localization

• Representation Structure

Mechanistic Interpretability

• Circuit Analysis

• Algorithm Discovery

• Causal Analysis

• Feature Analysis

Neuroscience-Inspired Interpretability

• Representational Similarity

• Dynamical Systems Analysis

• Population Coding

• Computational Neuroscience Comparisons

III. Mechanistic Interpretability Within the AI Landscape

Relationship to Capability Research

• Understanding Learned Computation

• Not Primarily Focused on Improving Performance

Relationship to Alignment Research

• Transparency

• Monitoring

• Safety Applications

Relationship to AI Science

• Reverse Engineering Neural Networks

• Discovering Computational Structures

• Explaining Model Behavior Mechanistically

Major Contemporary Schools

• Circuit-Centric Approaches: Transformer Circuits, Reverse Engineering Algorithms, Computational Graph Analysis

• Feature-Centric Approaches: Sparse Autoencoders, Feature Dictionaries, Neural Ontology Mapping

IV. Central Question of Mechanistic Interpretability

Core Research Agenda

- Identification of Internal Computations
- Discovery of Learned Algorithms
- Mapping of Representational Structures
- Explanation of Model Behavior Through Internal Mechanisms
- Understanding What Computational Structures Gradient Descent Creates